The Consequences of Poor Design: A Critique of the Systems Experiment

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Preface

While I was writing this essay, Turtle approached me and asked me to take over the operation of the Systems Experiment. I accepted, as I want to see the IMS and this event succeed. This is, of course, a potential conflict of interest, but I chose to disclose this information here because it is relevant to the context of this critique. For this reason, I also ask that you scrutinize this essay under that context. Though, I will not alter the language or any of my critiques in the essay to accommodate this change. Will I implement any significant measures in the event to offset the problems I bring up? Probably not. I'll try to communicate more to the best of my ability, but at this point in the event, it would be actively counterproductive and would also go against what I argue for. However, I welcome and will encourage people to criticize how I then run it.

Minecraft Server Theory arguably began with someone who actually put their theory into practice. PineconeLP's "Raid Farms" has been discussed at length in the IMS, whether in passing or as the focus of an entire essay or video—there's no doubt that it's influential. However, having only one example of a server using MST in its design becomes quite problematic in the long term for the field. As a result of this, there's a large amount of people in the IMS who've expressed wanting to experiment more, thereby putting more ideas into practice. This is difficult, of course, because running a server takes a lot of time and money, time and money that people don't have. And that's not even taking into account whether or not people want to run control studies or get studies peer-reviewed. For this reason, there

have not been many experiments.

The Systems Experiment is an event run by Turtle (turtle2770), an administrator at the IMS, who set out to create a space to run experiments. Unfortunately, the event fails at this goal. What could've been an environment, where members of the IMS could test their systems, was not. This essay is a look into what exactly went wrong and an urge to do right in the future. In writing this critique, I acknowledge that Turtle has a lot on his plate at the moment, and I sympathize with him as someone who often plans events myself. This is not a criticism directed at Turtle or any of the other organizers of the event; I have immense respect for their efforts. The Systems Experiment is an example of a well-intentioned, however, very flawed attempt at experimental research—flaws caused by its lack of adequate communication, ad hoc decision making, and lack of scientific integrity.

Effective communication is the foundation of any good event, and really any sort of social activity. Communication can make or break the success of an event. And it's unfortunately one of the biggest flaws of the Systems Experiment. Outlining well-defined goals is a prerequisite for effective communication. While the goals of the event were described in the Manual and Handbook, most people seemingly hadn't read them; indicated by the lack of a common sentiment held by the participants and certainly not one that most people understood. According to the Manual, the goal of the event was to make an environment fit for people to create and test systems in a vanilla setting while offering a competitive aspect ("Full Manual"). The lack of a clear goal in the eyes of the participants is indicative of the inadequate communication by event organizers, which then leads to a disconnect between the participants and organizers, hurting the overall event experience.

Along with this, there were scenarios during the event where the lack of communication from event organizers actively hurt my own team's "systems." This mostly revolved around the mechanics of the event, mods, rules, etc. One of my teammates' systems heavily relied on the number of Nether portals allowed per team. We received unclear communication on this issue so much so that it was contradictory at times. The changes to certain mechanics were often hidden behind a thread on the suggestion or question board. An example of this is how my team, *Ordonia*, wasn't aware of the wiretaps mod, while team *Alpha* was. Moreover, there were mods that neither team was made aware of. Inadequate communication hurts the results of the experiment, both in that inconsistent knowledge leads to inconsistent participants, i.e. uncontrolled participants, and in that it makes it hard for people that are trying to learn from the event to understand the parameters and context of the

Systems Experiment.

The lack of communication, however, is a symptom of another problem. Since the inception of the event, it has undergone many changes. It's not certain how many were due to participants' suggestions and how many were merely the result of an organizer's change in mind, but both of these avenues played a role. These changes were a misstep in the running of the event—they were often impromptu and unplanned while also being counterproductive to the goals of the experiment. To be clear, this is not coming from the angle of, "all change is bad." A lot of the changes made by Turtle were good, and as I've outlined in my essay, "The Reactionary Nature of Popular Answers to the Two Week Phase," anyone who adopts such an approach is mistaken. However, in this essay, we will consider the changes from the perspective of experiment design as well as participant experience where we find that they are counterproductive in achieving the goals of the experiment.

While ad hoc decision making may sound like not the worst thing, or perhaps, it may even sound positive—what's wrong with adaptability after all? For an experiment, it can be extremely detrimental. Changes introduce new variables into the equation, variables that weren't in existence before. These new variables render any data in between the changes to be in different contexts. Assumptions made with previous data must be reevaluated in a new context. This ad hoc decision making is further fueled by the "Suggestions" board on the Systems Experiment Discord server. Allowing players to have an input is usually a good thing. It allows players to build what they want to play and deepens a sense of community. However, in an experiment allowing participants to significantly affect the experiment creates some problems. For one, obvious reason: It affects the outcome of the study, precluding scientific testing of a hypothesis. Even if your participants are running their own "studies," these changes affect everyone, not just the participants who want the change. Not to mention that players are participants in their own experiments, which is opening up a whole other can of worms we will get to later. For these reasons, on the spot decisions just don't work if you want to yield reliable results.

On the player side of things, changes can foil any plans participants try to make. Especially in an event like this where planning is the name of the game, unpredictable changes can introduce problems that simply cannot be prepared for. Bringing us back to the Nether portal example. The rule was debated frequently, not only by participants but by organizers as well. The decision was only finalized nearly *a week* into the event. This meant a week of uncertainty on whether my teammates' system would actually work. The uncertainty

surely has knock-on effects (i.e., frustration and burnout), resulting in the loss of players the event desperately needs. These effects are antithetical to the goals of the event.

The frequent, ad hoc changes seen above suggest a lack of foresight and planning in the design of the Systems Experiment. It's understandable to want to meet the needs of the players, however, it comes at the cost of procedural control, a price that cannot be afforded in a situation like this. In that way, the Systems Experiment is sacrificing its ability to offer reliable results, compromising by pandering to participants' own sensibilities. Ironically, these ad hoc choices also made participant experience worse; creating an environment where it's hard to plan or even enjoy the event at all.

While the name of the event is the Systems *Experiment*, Turtle has stated that it is not an "experiment," and it was named the way it was for "branding" purposes (turtle2770. 2026A, 2026b). Even if we suppose that its goal is to allow people to experiment with new systems, it blunders. You'll frequently come across decisions that seem counterintuitive to achieving Systems Experiment's goals. In the "Design Mission" section of the manual, it states that one of the event's goals is to create systems that could "realistically be used by players in almost any server" ("Full Manual," 1). This suggests a desire to be able to generalize the results of experiments to areas outside of the "lab" (turtle2770. 2026c). While this goal is stated, it is not seriously taken into consideration.

One of the most glaring of these problems is having the designers of the systems be participants in *their own* experiments. It should be pretty obvious why this is a bad idea. The designers of the systems design them the way they do because they think it may work, hoping for it to succeed. This is clearly a conflict of interest. Consciously or not, the system's designers will affect the outcome of the system in ways that would not otherwise be seen. There are scenarios in research where you may want to have a researcher "participate" but only as a perceived peer by other participants which can be controlled (this kind of researcher is called a "confederate").

Further problems are found within the very nature of the event itself. The event, being a competition, encourages using and sticking with systems that might otherwise be faulty in the name of winning a competition. It creates a scenario where the players have an investment in keeping systems alive beyond their efficacy and material benefits. While this may represent some servers, it certainly does not represent most. It also discourages potentially malicious or otherwise self-interested players from interacting with a system in a way that is unexpected (i.e., a loophole). Even if such players are annoying, they are

extremely important to understanding a natural server's environment.

Not only does the Systems Experiment present a conflict of interest in how it is structured. This kind of attitude toward non-invested players is enshrined explicitly into the rules of the experiment. Later in the manual, it describes the rules of the event, one of them being that "a player is not allowed to try to purposefully evade the mechanics of a system," ("Full Manual," 10). A page below this rule, on allowing people to go without designing a system, states, "while it sounds fine in theory, this will introduce a lot of members who want to focus specifically on SMP style play. They are likely to get annoyed with systems over time and bypass them," ("Full Manual," 10). These rules may be well-intentioned, however, like with the structure of the event, they introduce a biased sample and a skewed dataset towards systems-oriented players. These rules are diametrically opposed to the goal of creating systems that could "realistically be used," ("Full Manual" 1).

Another sampling bias we see is due to the large amount of IMS members who joined the event. IMS members are interested in server design, and as a result of participants primarily being interested in server design, they consciously decide not to engage with certain features they deemed incompatible with the goals of the server. One of my teammates proposed that we went without using one of the mods used by the server entirely. This does not reflect natural player behavior that is necessary in experimentation; the average player is far more interested in achieving their own goals than in adhering to an idealistic purism based on a niche online community. This problem could be waved off as a failure in advertising. However, it connects to the issue of not allowing non-systems-oriented players into the event as well. The barring of non-design interested players from participating, whether indirectly or not, is detrimental to the usefulness of the data and therefore detrimental to experimentation.

Despite it being *the* foundational goal of the Systems Experiment, the event wasn't able to offer a testing ground that yielded generalizable conclusions. Thus, the "experiment" has failed to progress the field of MST. This is demonstrated by the biased effects that come from designers participating in their own systems, the nature of competition, and the exclusionary rules and invitations into the event. It's an example of a lack of scientific integrity which should be learned from for the future.

In this critique, we've seen a contradiction between the goal of a fun gamified competition (i.e. a good event) and allowing participants to test their own systems in a controlled environment (i.e. a good experiment). This contradiction ends up hurting both goals in the end. The event struggled to communicate with participants effectively while

trying to listen to as much as possible. Decisions were made in the moment, hurting the research value of the experiment, while claiming the goal of creating a space for experimentation. Furthermore, the design itself discouraged reliable and generalizable results that can be applied to many different servers. Don't take this as a call to change the Systems Experiment. As I've explained above, that would likely hurt the experiment even more. Let the Systems Experiment, my and hopefully others' critiques of it, be a learning experience for the IMS in its forthcoming scientific endeavors. Let it pave a new path forward for Minecraft Server Theory.

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